



CS321: Object Oriented Programming (2)

Final Project Report Template

Project Group Number (As In Blackboard):¹

Section Number:12

Project Title:event booking app



Members:

	Name	ID	Role Participated in	Sign
1	Zahiah alkhaldi(Leader)	2230004770	1. Login & Home Interface	✓
2	Lamia alqahtani	2230007219	2. Event Management	
3	Rahaf aldossari	2230003155	3. Movie & Exhibition Management	
4	Zainab alhashim	2230002555	4. Seat Selection & Booking	
5	Zainab almajed	2230004150	5.Payment & Feedback	
6	Sara ibnshamsah	2230003371	6. User Management	
7	Atheer aldighrir	2230000262	7. Database & Event Display	

¹ . Checklist Of Requirements Fulfilled In The Project:



Function / Application Aspect	Classes Management	Data Export/Import	Professional Appearance	Error Handling	Data Access Tools and Techniques
User Login	✓			✓	✓
Event Creation (Admin)	✓			✓	✓
Event Browsing (Customer) Seat Selection & Booking	✓			✓	✓
Payment Process	✓			✓	✓
Feedback Submission	✓	✓		✓	✓
Image Upload & Preview	✓	✓		✓	✓
Search by Event ID	✓	✓		✓	✓



2. Difficulties & How Problems Were Solved:

Database Connection Issues: Initially, we encountered issues with connecting the Java application to MySQL. This was resolved by configuring the JDBC driver (mysqlconnector-j-9.2.0.jar) properly and ensuring correct credentials and URL format

image Upload & Display: Integrating image uploading with proper preview was tricky. We solved it using ImageIcon and getScaledInstance() to adjust images dynamically.

Class Hierarchy Design: It was challenging to organize inheritance between event types (Movie, Festival, etc.). We resolved this by using a base Event class and creating specific subclasses for each event type

3. How Were the Tasks Distributed Equally Between Members?

The project was divided into 7 distinct functional modules, and each student was responsible for one module based on their skill level and interests:

Responsibility	Student
Login & Home Interface – Login validation and navigation design	Zahiah alkhalidi
Event Management – Event.java, EventManager.java	Lamia alqahtani
Movie & Exhibition Management – Movie.java, Exhibition.java	Rahaf aldossari
Seat Selection & Booking – SeatSelectionScreen.java, seat reservation logic	Zainab alhashim
Payment & Feedback – payment.java, Feedback.java, file export	Zainab almajed
User Management – User.java, Customer.java, role handling	Sara ibnshamsah
Database & Event Display – DatabaseConnection.java, EventBrowse.java	Atheer aldighrir

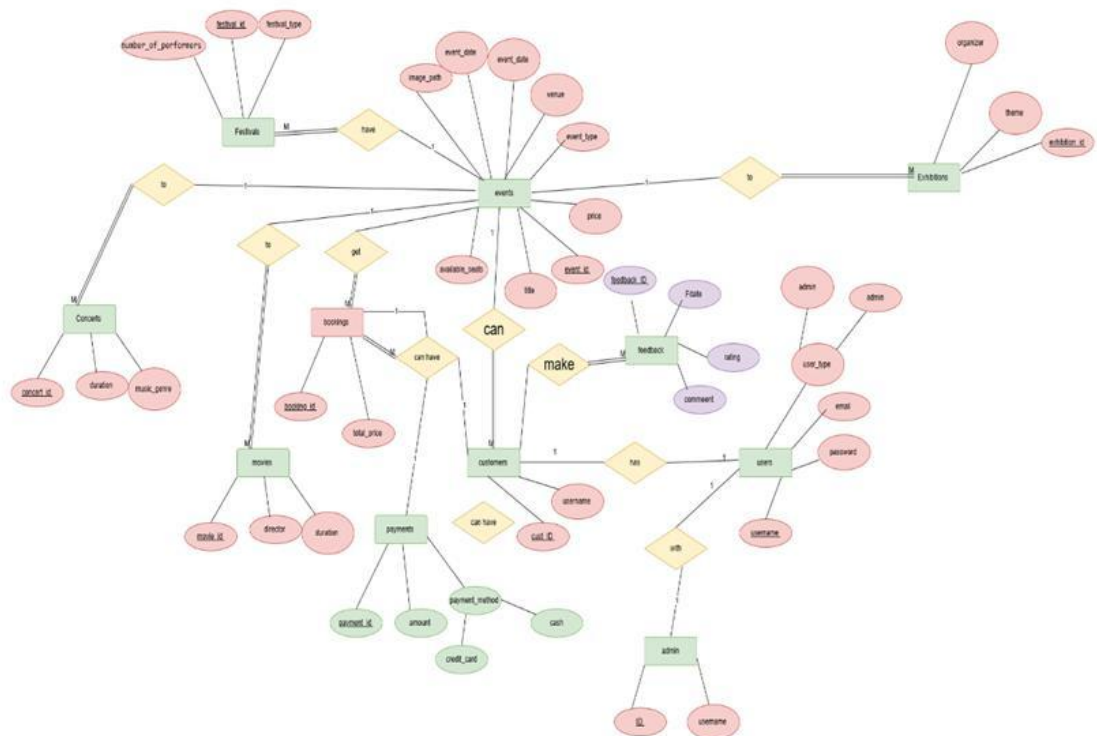
4. General Comments:

5. Attachments:

[https://drive.google.com/drive/folders/1jVGW5o13iTSwFCzYmU9WAngkMCXJmFB?usp=drive link](https://drive.google.com/drive/folders/1jVGW5o13iTSwFCzYmU9WAngkMCXJmFB?usp=drive_link)

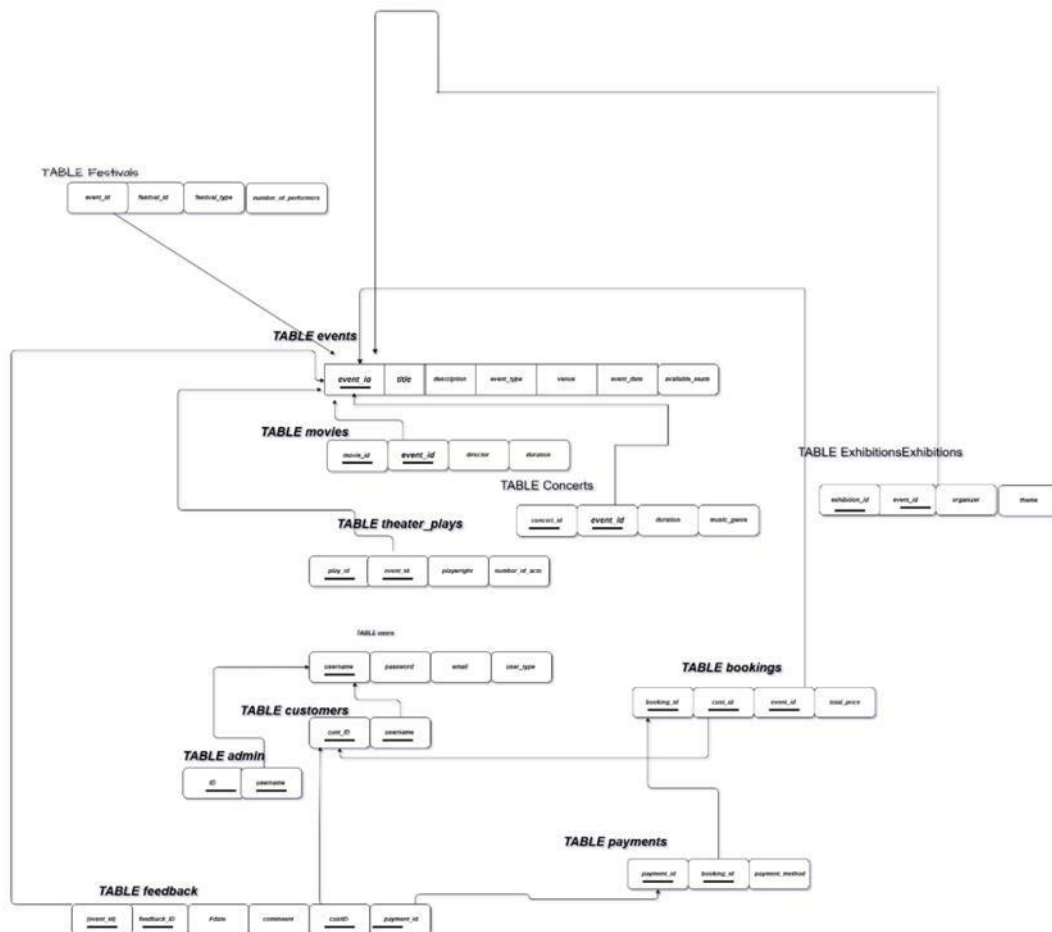


1.Er Design





2.Database schema



3.Data Description:

Users:

- username (Primary Key)/VARCHAR(50) - Unique username for login



- password/VARCHAR(255) - Encrypted password for security
- email/VARCHAR(100) - Unique email address for communication
- user_type/ENUM('admin','customer') - User role with default

'customer' **Customers:**

- cust_ID (Primary Key)/INT - Auto-incremented customer ID
- username/VARCHAR(120) - References users.username as foreign

key **Admin:**

- ID (Primary Key)/INT - Auto-incremented admin ID
- username/VARCHAR(120) - References users.username as foreign

key **Events:**

- event_id (Primary Key)/INT - Auto-incremented event ID
- title/VARCHAR(100) - Name/title of the event
- description/TEXT - Detailed description of the event
- event_type/VARCHAR(50) - Type/category of event
- venue/VARCHAR(100) - Location where event is held
- event_date/DATETIME - Date and time of the event
- image_path/VARCHAR(255) - Path to event image/photo
- price/DECIMAL(10,2) - Ticket price for the event
- available_seats/INT - Number of available seats/tickets

Movies:

- movie_id (Primary Key)/INT - Auto-incremented movie ID
- event_id/INT - References events.event_id as foreign key
- director/VARCHAR(255) - Name of movie director
- duration/INT - Movie length in minutes

Concerts:

- concert_id (Primary Key)/INT - Auto-incremented concert ID
- event_id/INT - References events.event_id as foreign key
- duration/INT - Concert length in minutes
- music_genre/VARCHAR(50) - Genre of music (rock, pop, etc.)

Bookings:

- booking_id/INT(11) - Booking reference number
- event_id/INT(11) - References the booked event
- seat_id/INT(11) - References the booked seat
- total_price/DECIMAL(10,2) - Total amount paid for bookin

Payments:

- payment_id (Primary Key)/INT - Auto-incremented payment ID
- amount/DECIMAL(10,2) - Payment amount
- payment_method/VARCHAR(50) - Payment type (credit card, cash,

etc.) **Feedback:**



- feedback_ID (Primary Key)/INT - Auto-incremented feedback ID
- Fdate/VARCHAR(120) - Date when feedback was submitted
- rating/INT - Rating score (e.g., 1-5 stars)
- comment/VARCHAR(120) - Customer comments/review
- custID/INT - References customers.cust_ID as foreign key

Exhibitions:

- exhibition_id (Primary Key)/INT - Auto-incremented exhibition ID
- event_id/INT - References events.event_id as foreign key
- theme/VARCHAR(100) - Exhibition theme/topic
- organizer/VARCHAR(100) - Organization hosting the exhibition

Festivals:

- festival_id (Primary Key)/INT - Auto-incremented festival ID
- event_id/INT - References events.event_id as foreign key
- festival_type/VARCHAR(100) - Type of festival (music, food, etc.)
- number_of_performers/INT - Count of performers/participants

Seats:

- seat_id/INT(11) - Seat identification number
- event_id/INT(11) - References events.event_id as foreign key
- seat_number/VARCHAR(10) - Seat number (e.g., "A12")
- row_number/VARCHAR(5) - Row number/letter
- section/VARCHAR(20) - Section/area of venue
- status/ENUM('available','booked','reserved') - Current seat status
- price/DECIMAL(10,2) - Price for this specific seat

4.database Tables with filled data :

User:

	username	password	email	user_type
▶	admin1	securepass123	admin1@example.com	admin
	admin2	adminpass456	admin2@events.com	admin
	ali_ahmed	ali5678	ali.ahmed@example.com	customer
	event_lover	events123	eventfan@example.com	customer
	event_manager	managerpass	manager@events.com	admin
	fatima_z	fatima2023	fatima.z@example.com	customer
	jane_smith	janespass456	jane.smith@example.com	customer
	john_doe	johnspass123	john.doe@example.com	customer
	mohamed_k	mohamed2023	mohamed.k@example.com	customer
	movie_buff	cinema789	moviefan@example.com	customer
	music_fan	music456	musiclover@example.com	customer

Customer:



cust_ID	username
8	ali_ahmed
13	ali_ahmed
3	event_lover
9	fatima_z
14	fatima_z
2	jane_smith
1	john_doe
7	mohamed_k
12	mohamed_k
5	movie_buff
4	music_fan

Admin:

ID	username
1	admin1
2	admin2
3	event_manager
4	super_admin
NULL	NULL

Events:

event_id	title	description	event_type	venue	event_date	image_path	price	available_seat
1	Avengers: Endgame	The epic conclusion to the Avengers saga	movie	Grand Cinema	2023-06-15 19:00:00	avengers.jpg	15.99	200
2	Coldplay Concert	Music of the Spheres World Tour	concert	Stadium Arena	2023-07-20 20:30:00	coldplay.jpg	89.99	5000
3	Art Exhibition	Modern Art Masters Collection	exhibition	City Art Gallery	2023-06-10 10:00:00	art.jpg	12.50	100
4	Summer Music Festival	3-day outdoor music festival	festival	Riverside Park	2023-08-05 12:00:00	festival.jpg	149.99	10000
5	Inception	Christopher Nolan masterpiece	movie	Classic Theater	2023-06-22 18:00:00	inception.jpg	12.99	150
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Booking:

booking_id	event_id	seat_id	total_price
1	1	1	25.99
2	2	4	99.99
3	3	6	12.50

FeedBack;

feedback_ID	Fdate	rating	comment	custID
1	2023-05-15	5	Amazing event! Will come again.	1
2	2023-05-16	4	Great experience, but seats were a bit cramped.	2
3	2023-05-17	3	Good event but overpriced.	3
NULL	NULL	NULL	NULL	NULL



Seats:

	seat_id	event_id	seat_number	rownumber	section	status	price
▶	1	1	A1	A	Premium	available	25.99
	2	1	A2	A	Premium	available	25.99
	3	1	B1	B	Standard	available	15.99
	4	2	101	1	Floor	available	99.99
	5	2	102	1	Floor	available	99.99
	6	3	G1	G	General	available	12.50
	7	4	FEST1	1	GA	available	149.99
	8	5	C3	C	Standard	available	12.99

Exhibitions:

	exhibition_id	event_id	theme	organizer
▶	1	3	Modern Art Masters	City Arts Council
	2	3	Modern Art Masters	City Arts Council
*	NULL	NULL	NULL	NULL

Festivals:

	festival_id	event_id	festival_type	number_of_performers
▶	1	4	Music	25
	2	6	Music	35
	3	7	Food	50
	4	8	Film	20
	5	9	Cultural	75
*	NULL	NULL	NULL	NULL

Movies:

	movie_id	event_id	director	duration
▶	1	1	Anthony Russo, Joe Russo	181
	2	5	Christopher Nolan	148
*	NULL	NULL	NULL	NULL

Concerts:



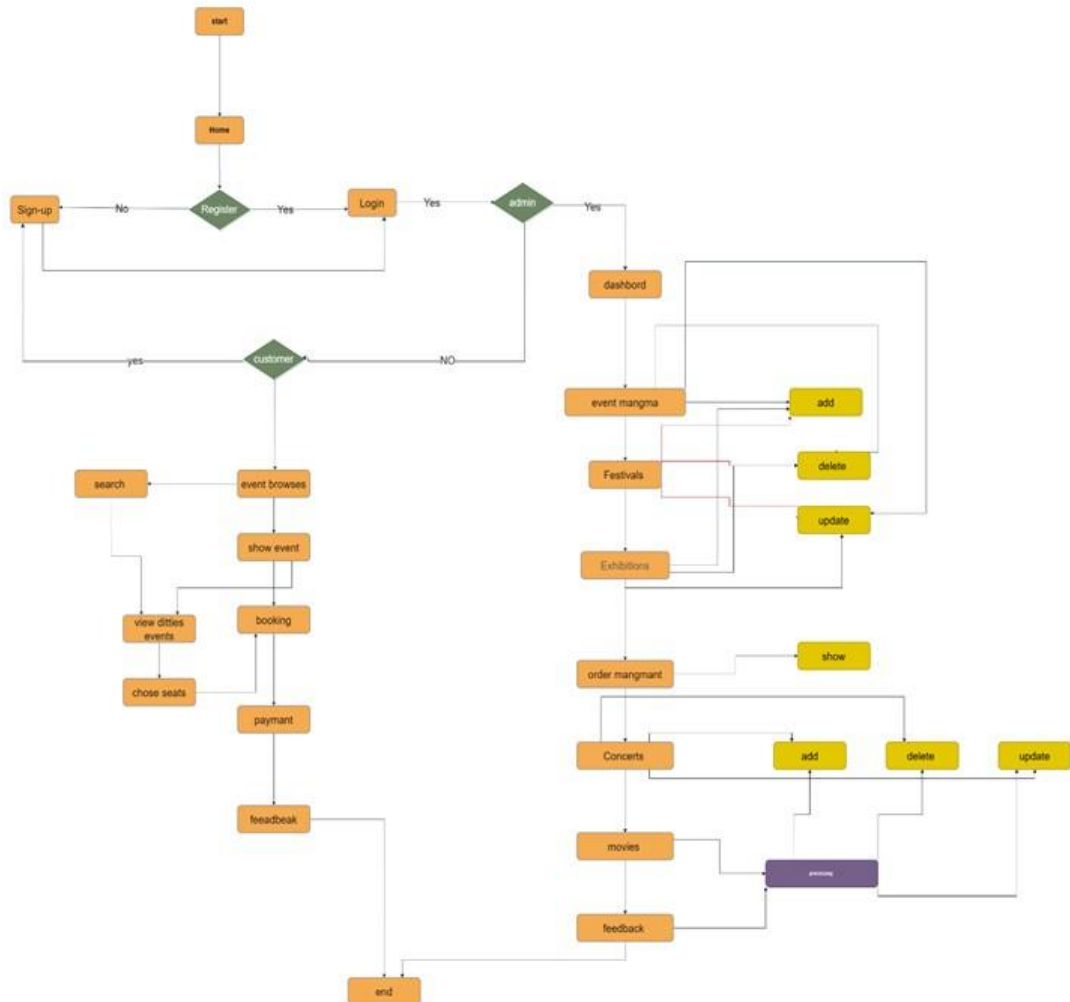
concert_id	event_id	duration	music_genre
1	2	120	Alternative Rock
2	2	120	Alternative Rock
NULL	NULL	NULL	NULL

Payments:

payment_id	amount	payment_method
1	25.99	Credit Card
2	99.99	PayPal
3	12.50	Credit Card
NULL	NULL	NULL

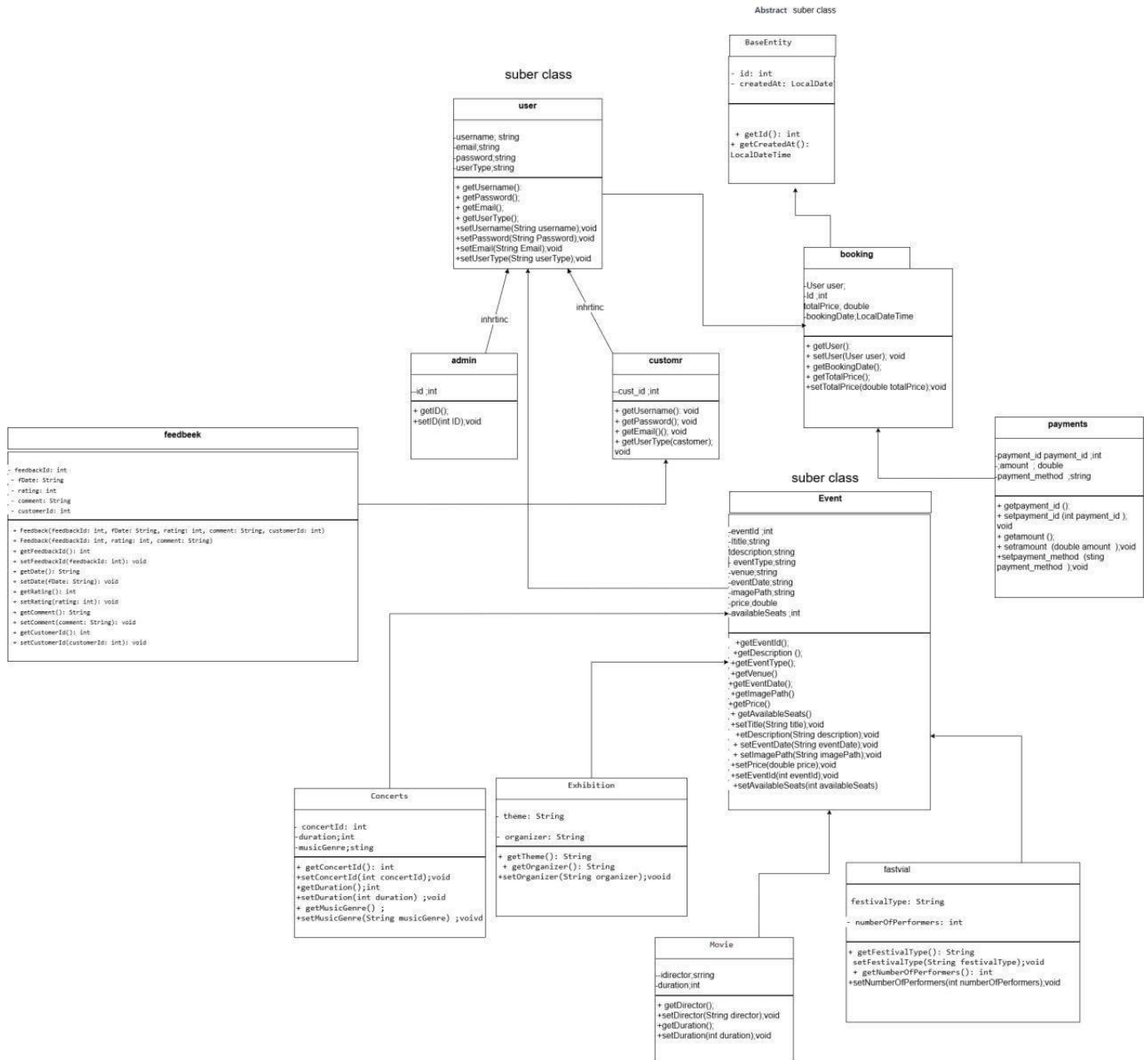


5.Applicaion Structure





6-UML DIGRAM



Application's Interface Design:

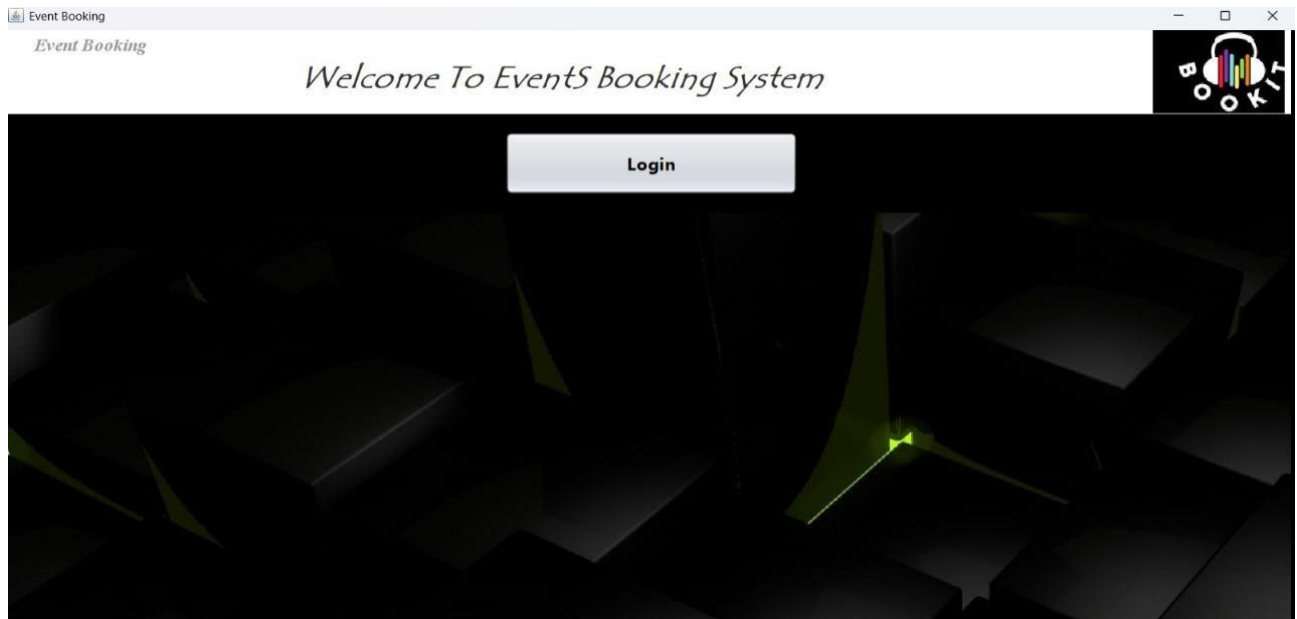


Figure1: Initial interface after choosing Login.

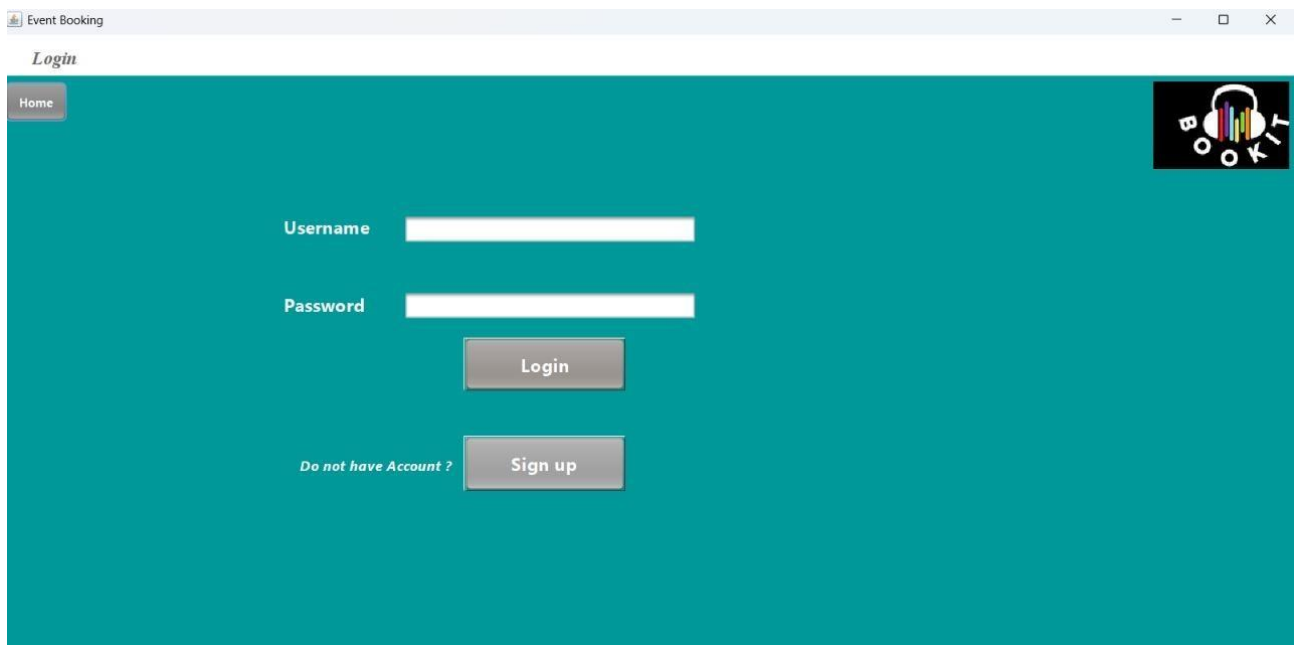


Figure2: inserting the username and password, to log into the account.

Event Booking

Sign up

Back

Username

Password

Email

User Type

Customer

Customer

Admin

Create Account

Figure3: creating account using email and user type (Customer, Admin).

Event Booking

Sign up

Back

Username

hala

Password

Email

halamohammed0@gmail.com

User Type

Customer

Create Account

Message

User registered successfully!

OK

Figure4: Creating the account successfully.

Event Booking

Login

Home

Username:

Password:

Login

Do not have Account ? Sign up

BOOKIT

Figure5: login again into the account after creating it.

Browse Events

ID:

Title:

Description:

Type:

Venue:

Date:

Show Events

Price:

Available Seats:

Event Details

event_id	title	description	event_type	venue	event_date	image_path	price	available...
1	Captain A	As Steve	movie	Cinema H	2023-12-1	F\worksp	120.00	24
3	oop 2	project	Exhibition	Exhibition	2025-06-0	C:\Users\	30.00	150
4	Venom po	As Steve	movie	Cinema H	2023-12-1	F\worksp	85.00	150
5	fight	fight	movie	Cinema B	2025-06-0	F\worksp	50.00	500
7	Exhibition	khlof	Exhibition	Cinema B	2025-06-0	F\worksp	50.00	300
9	rice music	vrey good	Concert	Concert H	2025-06-0	C:\Users\	75.00	300
10	good man	good man	movie	Cinema Hall	2025-06-0	D\photos	50.00	200
15	captain A	capthahogv	movie	Cinema Hall	2025-06-0		50.00	200
16	Spider-Man	Peter Par	movie	Cinema H	2023-12-2	F\worksp	80.00	200
17	Spider-Man	Peter Par	movie	Cinema H	2023-12-2	F\worksp	80.00	200
18	Mockingjay	Katniss E	movie	Cinema H	2023-12-2	F\worksp	81.00	175
20	gimore girls	life of moth	Movie	Cinema Hall	2025-06-0		50.00	200
99	group 1	proect	Exhibition	Exhibition	2025-06-0		30.00	150
100	Captain A	As Steve	movie	Cinema H	2023-12-1	C:\Users\	120.00	24

BOOKIT

Figure6: The interface to book a movie and see the offers.

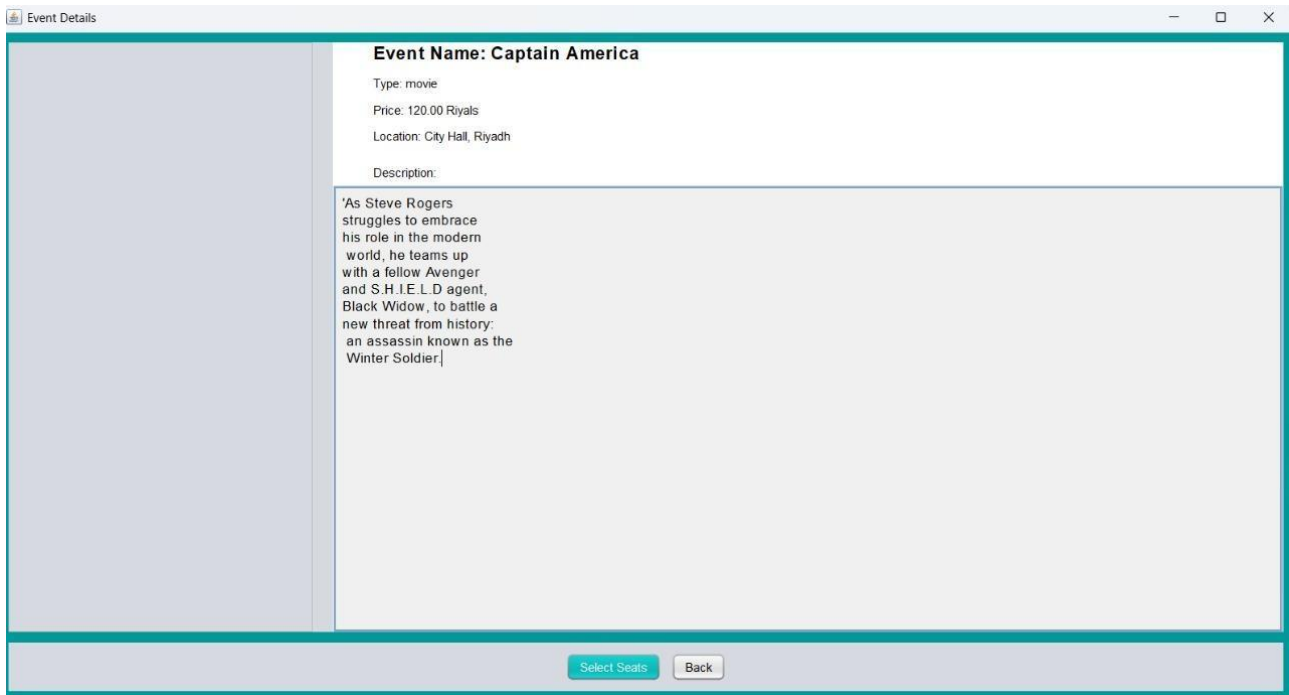


Figure7: This screen displays detailed information about the selected event before booking.



Figure8: This screen displays detailed information about the selected event before booking.



Select Seats - Event: Captain America

STAGE

B1	B2	B3	C1	C2	C3	A1	A2	A3			
----	----	----	----	----	----	----	----	----	--	--	--

Booking Confirmed

Booking Confirmed!
 You have successfully booked 2 seat(s)
Total Amount: 650.00 SAR
 Thank you for your purchase!

Seat Details

Selected Seat: C3
Section: Regular
Price: 150.00 SAR

Total: 650.00 SAR | Selected: 2 seats

☐ Available
 ☒ Selected
 ☐ Reserved
 ☐ Booked

Confirm Selection

Figure9: This screen confirms the user's seat selection and displays the total booking amount.

Payment

Event ID:

Total Amount:

Pay_Method: ☒ Credit Card ☐ Cash

Pay

Figure10: This screen allows the user to choose a payment method and confirm the total amount.



The screenshot shows a web application window titled "Payment". The main content area has a teal background. At the top, the word "Payment" is displayed in white. Below it, there are two input fields: "Event ID" with the value "1" and "Total Amount:" with the value "650.0". A modal dialog box is centered on the screen, titled "Message", with a blue information icon and the text "Successfull Payment". Below the modal, there is a "Pay" button. The bottom of the screen has a dark grey footer bar.

Figure11: This screen confirms that the

The screenshot shows a web application window titled "Rating". The main content area has a teal background. At the top, the word "Rating" is displayed in white. Below it, the text "How would you rate our servies ?" is shown. There are five radio buttons for rating, with the fifth button (5) selected. Below the rating, there is a text input field labeled "Any comment :" containing the word "great". A "Submit" button is located at the bottom of the form. The bottom of the screen has a dark grey footer bar.

Figure12: This screen allows the user to rate the service and submit a comment after completing the payment.

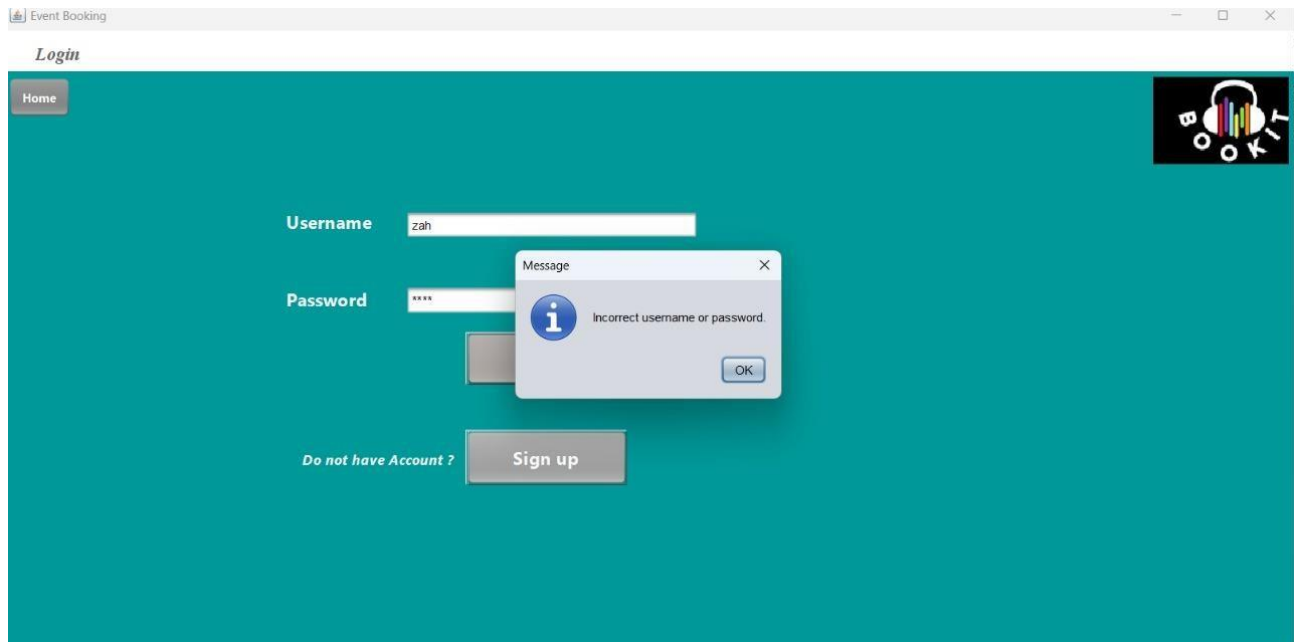
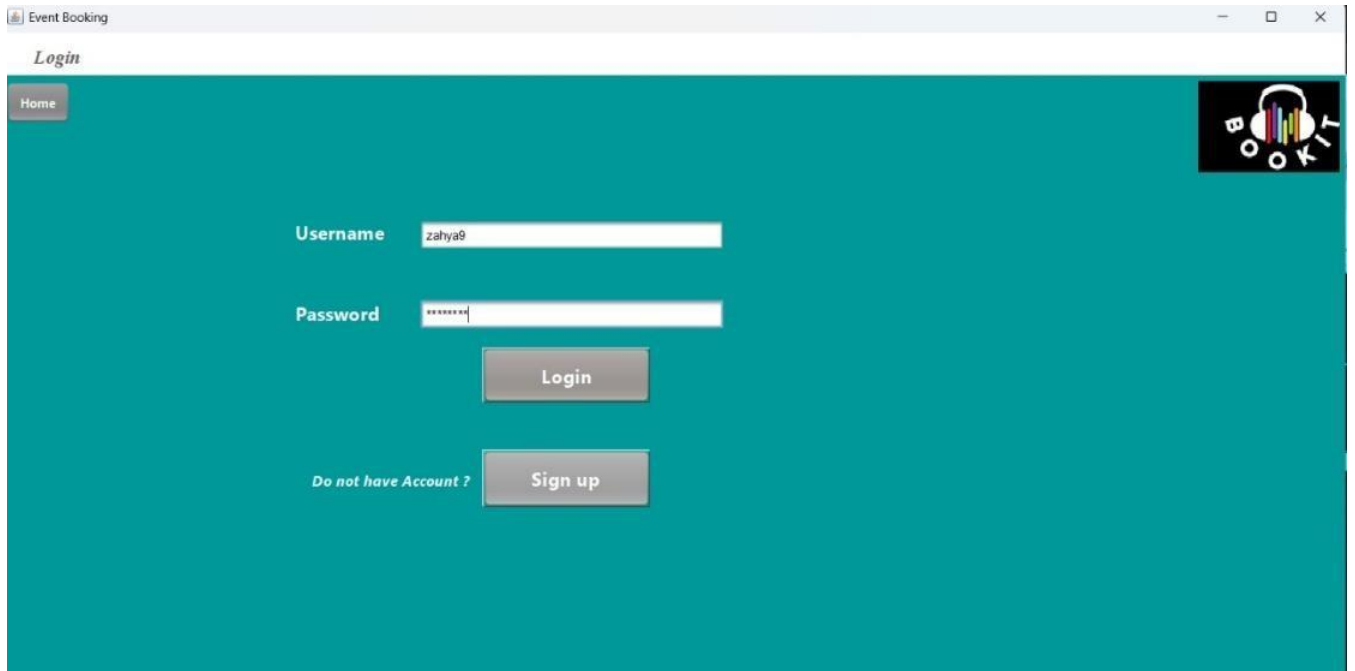


Figure 13: While creating a user who is not registered already.

Extra options

user_type	email	password	username	
customer	halamohammed0@gmail.com	Zz112233	hala	حذف - نسخ تعديل ☐
customer	h@gmail.com	123	hasan	حذف - نسخ تعديل ☐
customer	mmalkhaldi	Zz112233	mohammed	حذف - نسخ تعديل ☐
customer	zahiaalkhaldi7@gmail.com	Zz12345	zahya	حذف - نسخ تعديل ☐
admin	zaalkhaldi12@gmail.com	Zz11233	zahya5	حذف - نسخ تعديل ☐
admin	zozy2240@gmail.com	Zz112233	zahya9	حذف - نسخ تعديل ☐
customer	Zahyah@gmail.com	223	Zahyah	حذف - نسخ تعديل ☐
admin	zahiaalkhaldi7@gmail.com	Zz112233	zozy	حذف - نسخ تعديل ☐

Figure 14: Login from any Admin entry in the database.



Event Booking

Login

Home

Username

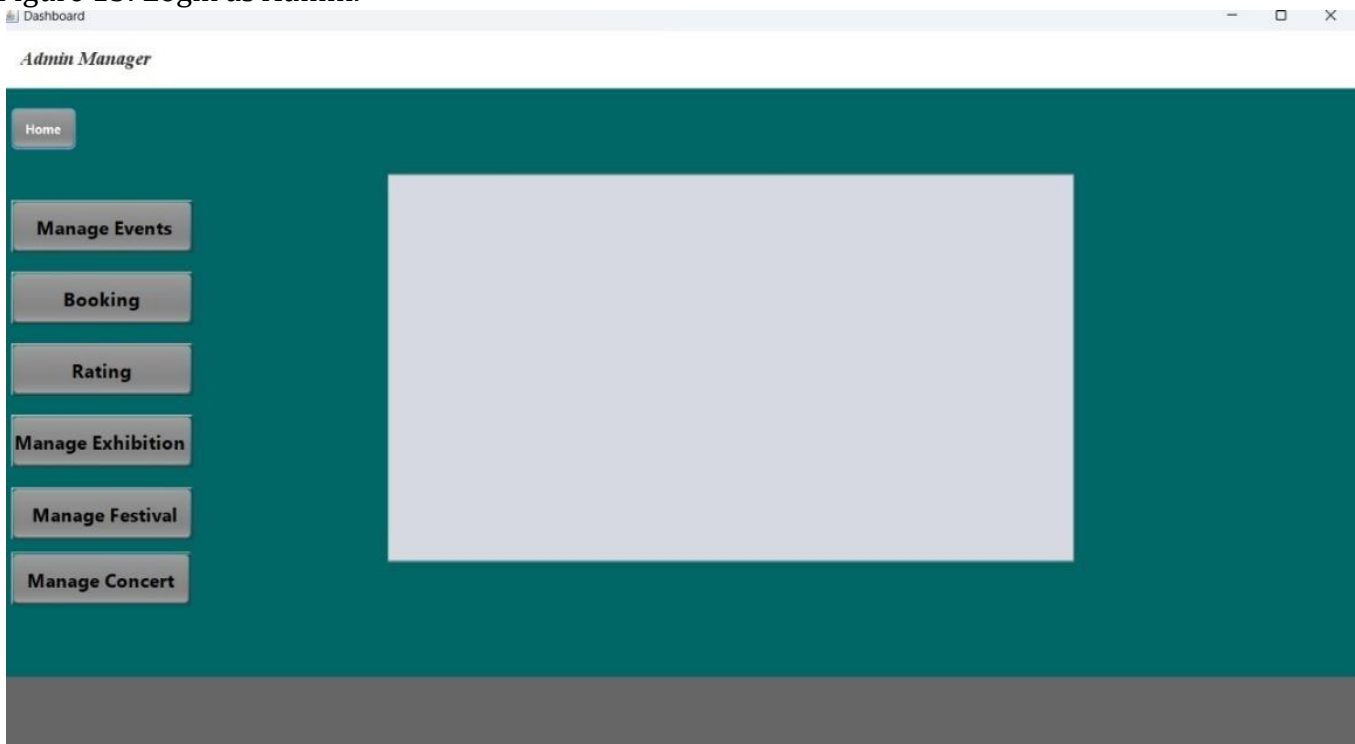
Password

Login

Do not have Account ?

BOOKIT

Figure 15: Login as Admin.



Dashboard

Admin Manager

Home

Manage Events

Booking

Rating

Manage Exhibition

Manage Festival

Manage Concert

Figure 16: Main Admin Interface.



Dashboard

Admin Manager

[Home](#)

[Manage Events](#)

[Booking](#)

[Rating](#)

[Manage Exhibition](#)

[Manage Festival](#)

[Manage Concert](#)

booking_id	event_id	seat_id	total_price
81	1	12	500.00
82	1	17	150.00
83	1	16	150.00
84	10	249	150.00
85	10	239	500.00
86	10	240	500.00
87	10	242	300.00
88	9	225	300.00
89	9	207	300.00
90	7	132	500.00
91	7	138	300.00
92	4	85	150.00
93	4	84	300.00
94	1	10	500.00
95	1	18	150.00

Figure 17: Under the 'Booking' Tab, we can see all selected events

ID : 1

Title spider man

Description: Black Widow, to battle a new threat from history, an assassin known as Winter Soldier.

Type movie

Venue Cinema Hall B

Date 2023-12-16 20:30:00

Image: 

Price 120.0

Available Seats 24

Add

Edit

Remove

ID	Title	Description	Type	Venue	Date	Image	Price	Available s...
1	Captain Am...	As Steve	movie	Cinema Ha...	2023-12-16	F:\works\pr...	120.0	24
4	Venom pos...	As Steve R...	movie	Cinema Ha...	2023-12-16	F:\works\pr...	85.0	150
5	ighly	ighly	movie	Cinema B	2025-06-01	F:\works\pr...	50.0	500
			ighly	Cinema B	2025-06-01	F:\works\pr...	50.0	300
			movie	Cinema Hall	2025-06-01	D:\photos\...	50.0	200
	ood &	Concert	Concert Hall	2025-06-01	D:\photos\...	75.0	300	
	man L	movie	Cinema Hall	2025-06-01	D:\photos\...	50.0	200	
		Concert	Concert Hall	2025-06-01		75.0	300	
15	captain Am...	cghvalicgv	movie	Cinema Hall	2025-06-01		50.0	200

Add Event

New Event Added Successfully

OK

Figure 18: When clicking 'Manage Event', the manager can add or update events

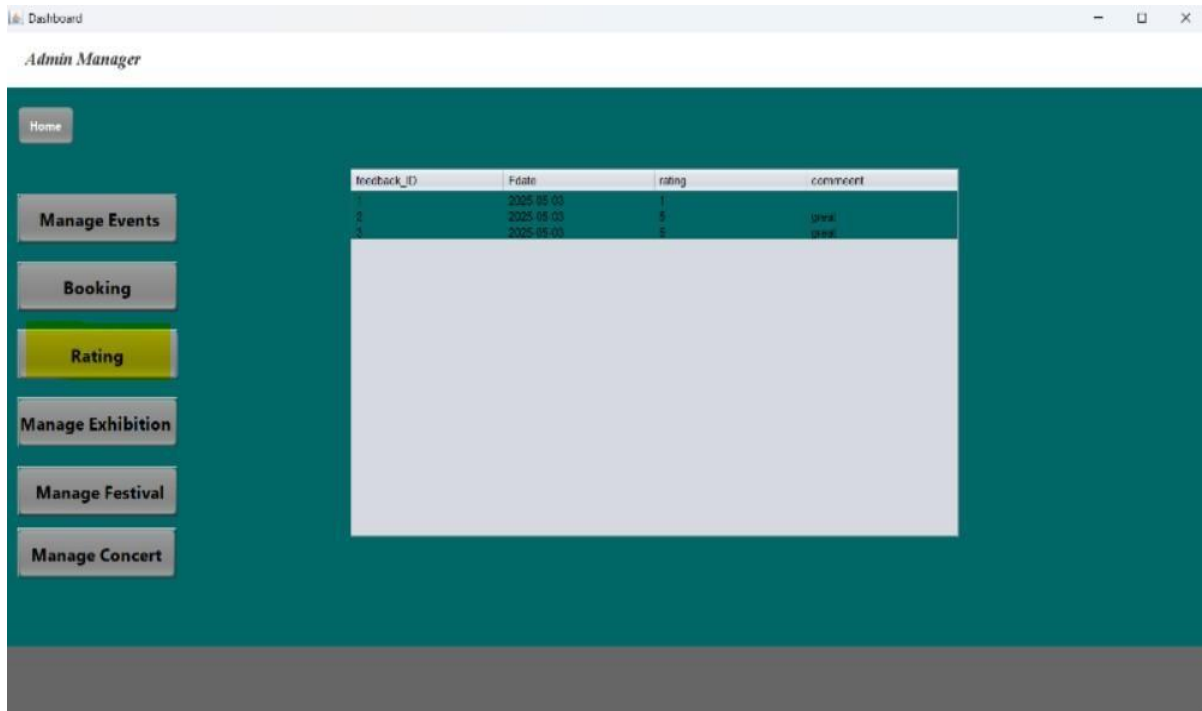


Figure 19: From 'Rating', it shows the date of feedback, the user's rate and his/her opinion

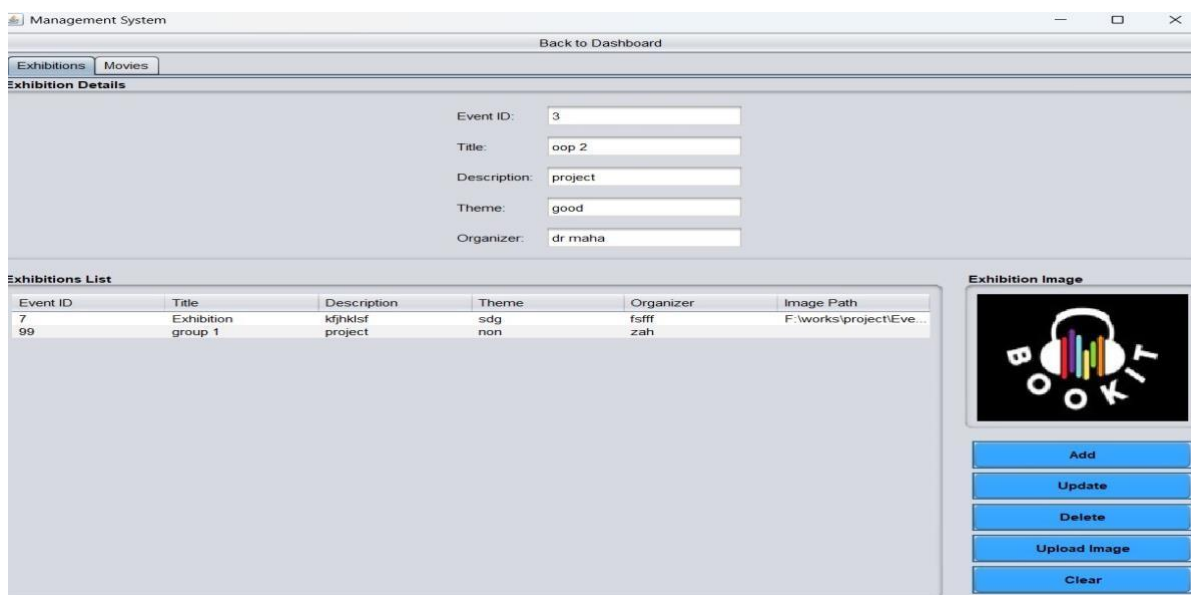


Figure 20: From 'manage exhibition', the manager can add, update, or delete an exhibition or movie

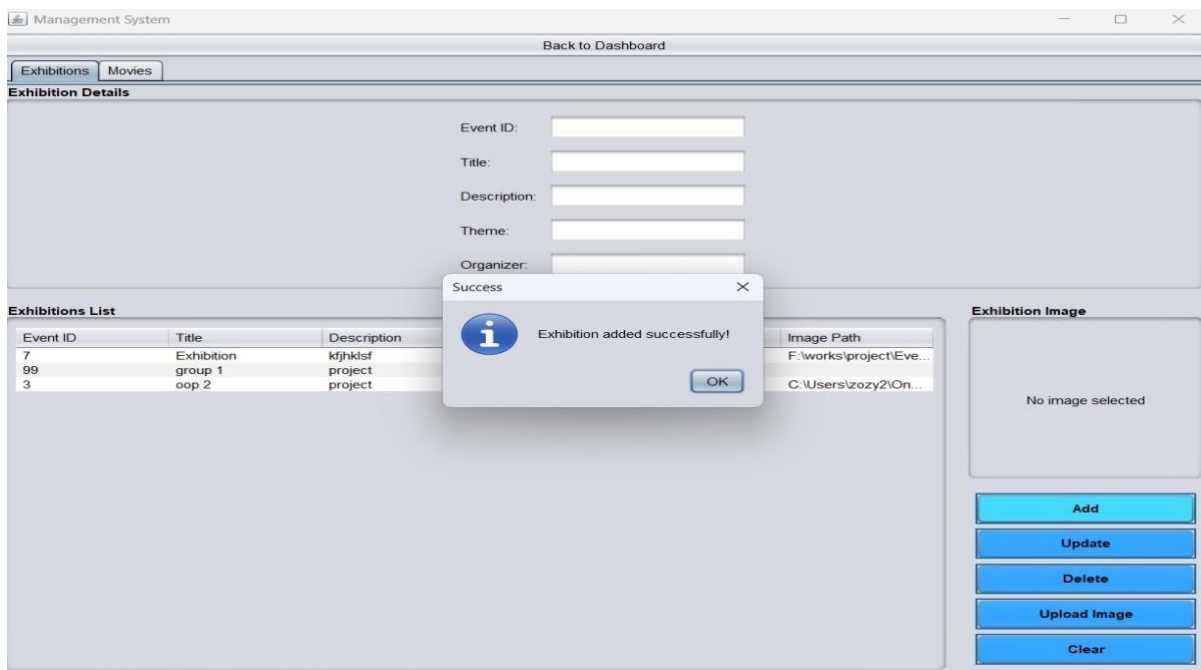


Figure 21: Adding an exhibition

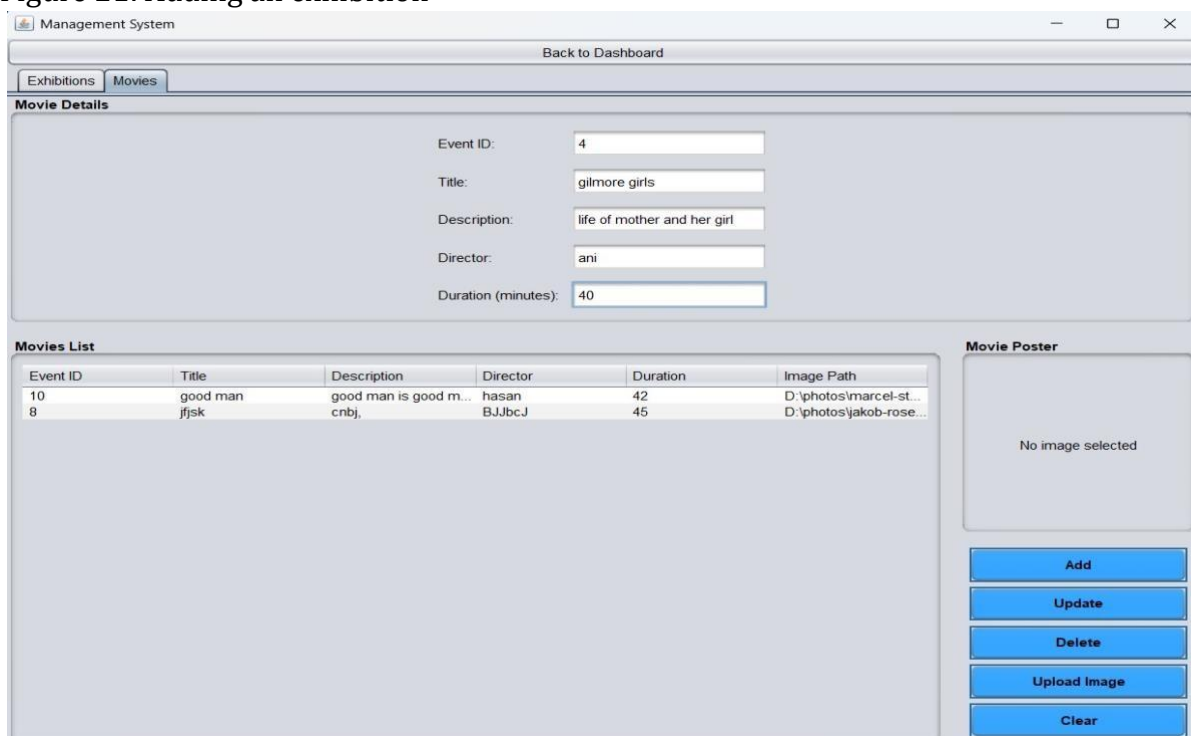


Figure 22: Adding a movie by writing name, id, description, director's name, and movie duration

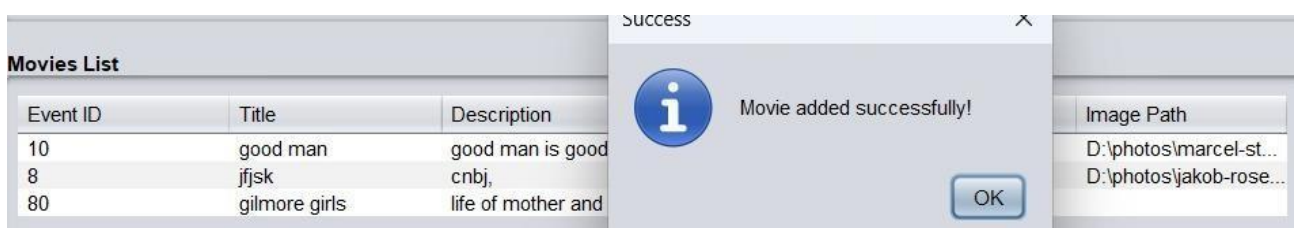


Figure 23: Movie Added

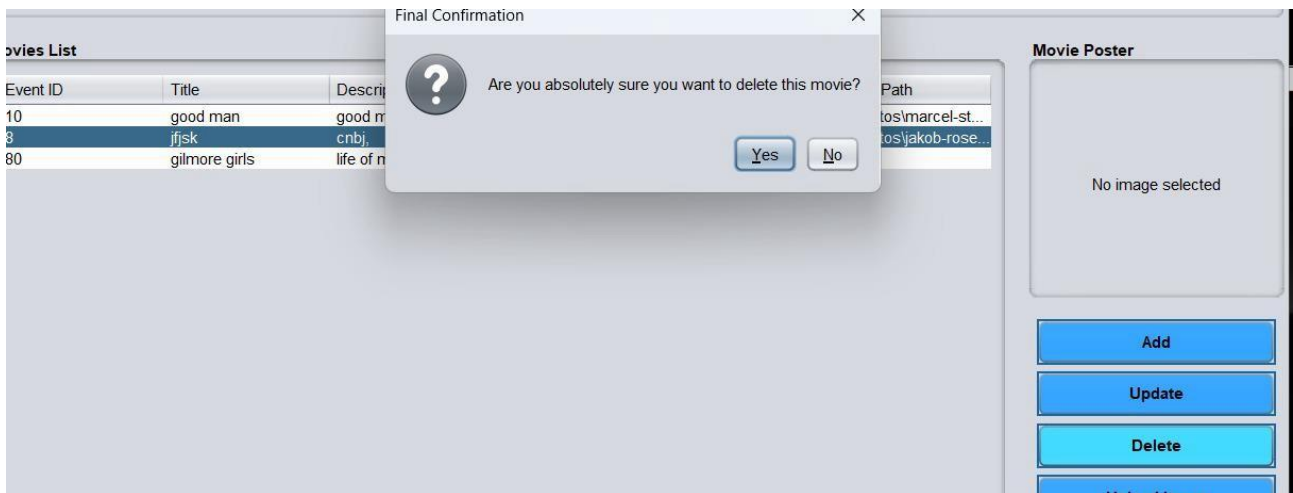


Figure 24: After selecting a movie to delete, the user is asked for confirmation.

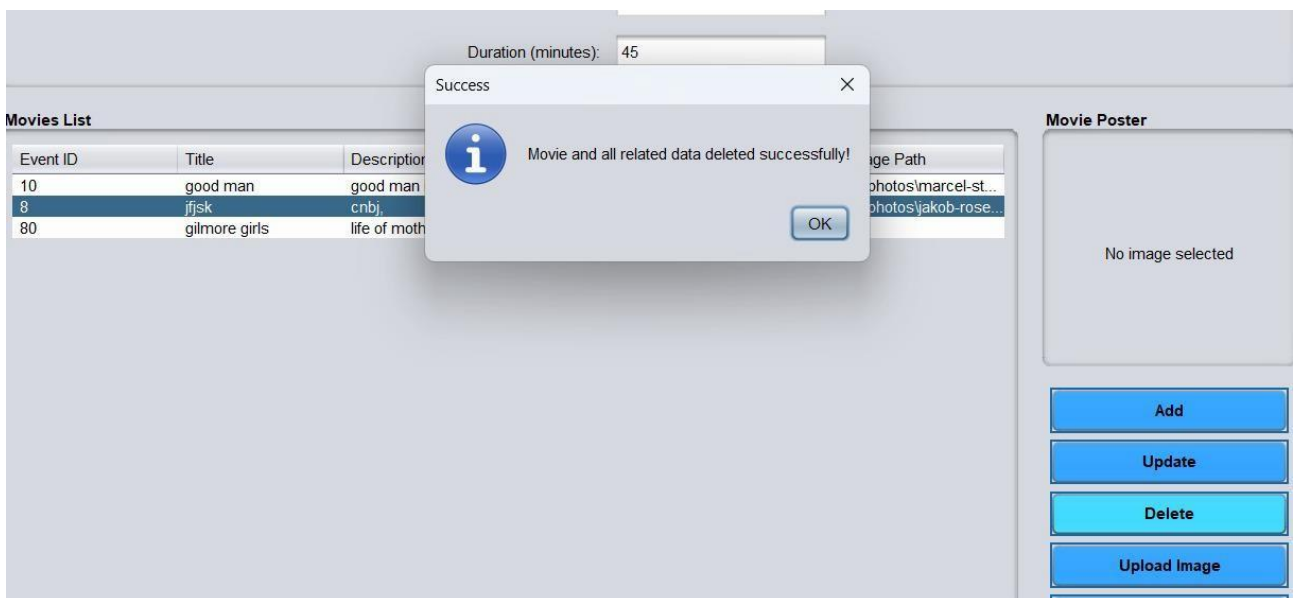


Figure 25: The movie and related data were deleted successfully.

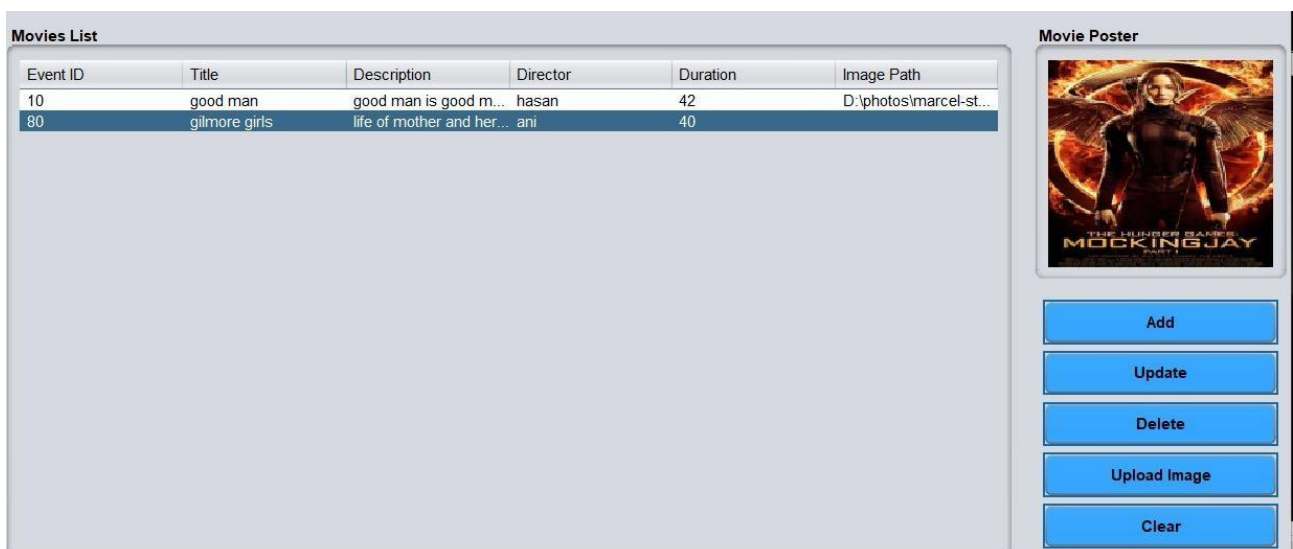




Figure 26: The list after deleting the movie.

Festival Manager

Event ID:

Title:

Description:

Festival Type:

Number of Performers:

Festivals List

Event ID	Title	Description	Festival Type	Performers	Image Path
5	fghhj	fghjk	jhgj	50	F:\works\projectE...

Festival Image

Figure 27: From festival: adding, deleting, or updating.

Concert Manager

Event ID:

Title:

Description:

Duration (minutes):

Music Genre:

Concerts List

Event ID	Title	Description	Duration	Music Genre	Image Path
9	nice music	vrey good lisetn	12	dance	D:\photos\william-d...
11	h	huhi	250	dancing	

Concert Image

Figure 28: From concert: adding, deleting, or updating.

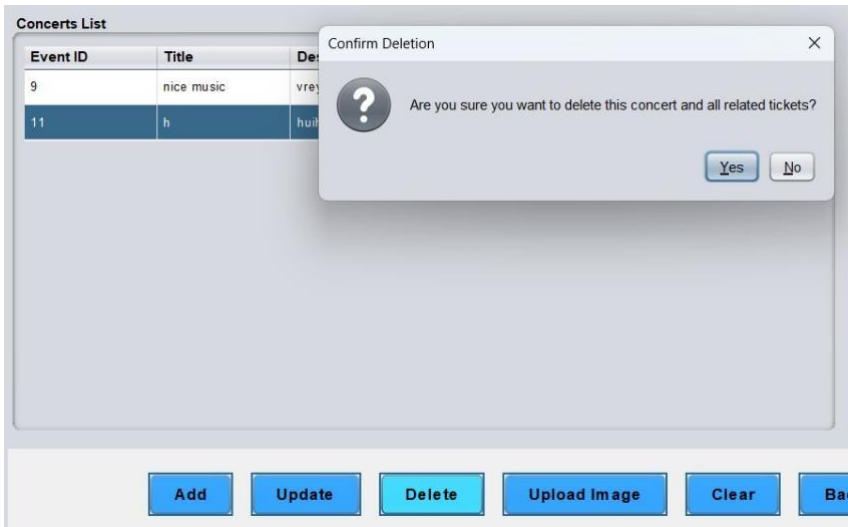


Figure 29: After selecting a concert to delete, the user is asked for confirmation.

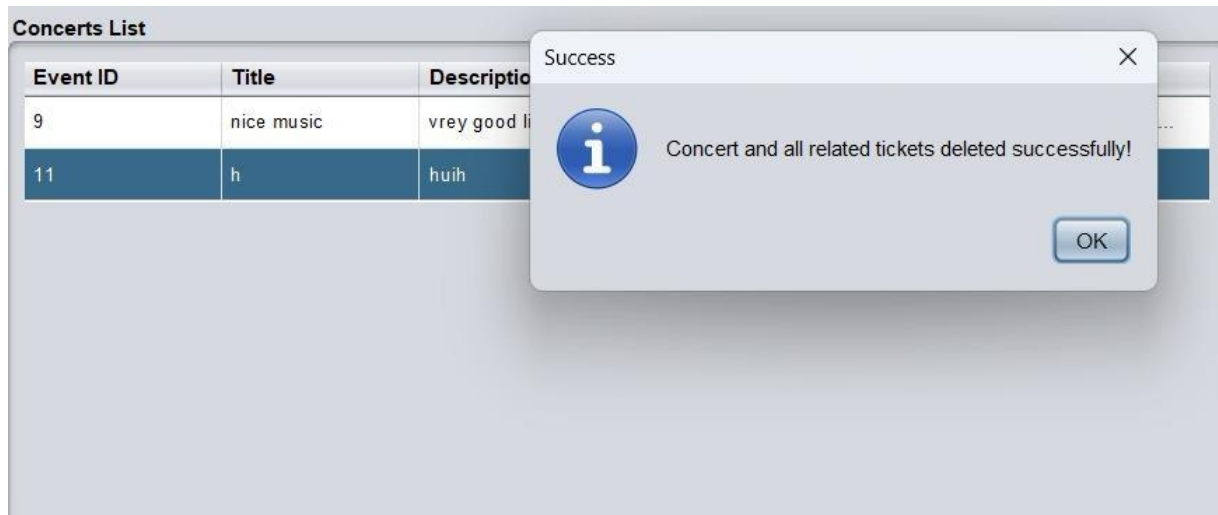


Figure 30: Confirming concert deletion.

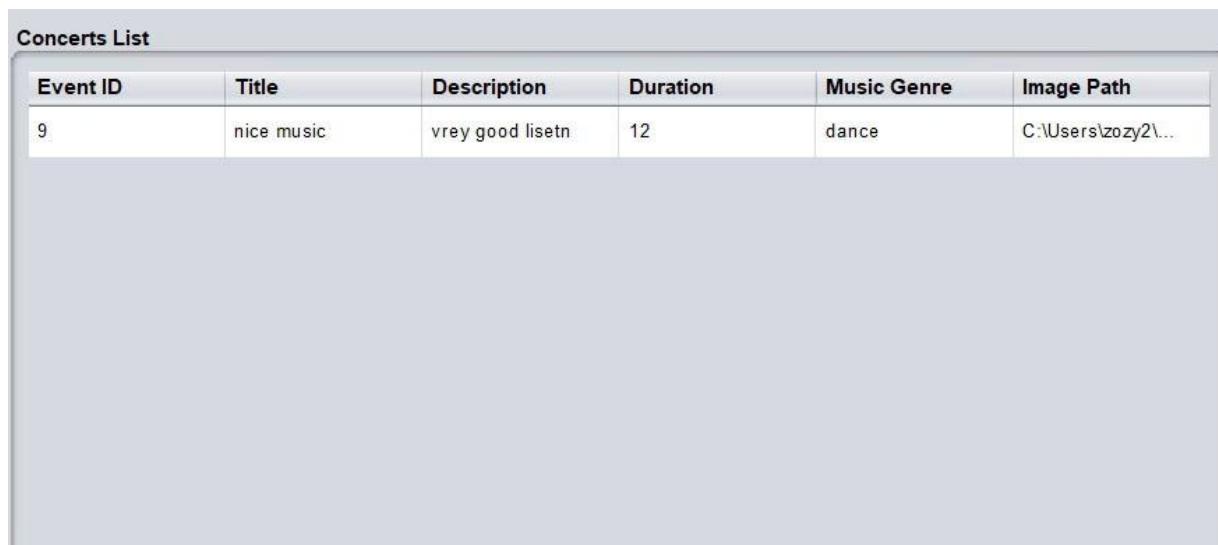


Figure 31: The list of concerts after deletion.